



End of Day Process

User Guide — EOD Snapshots, P&L Attribution & Reporting

MKM Research Labs

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MKM Research Labs

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1 Introduction

The End of Day (EOD) process is the daily checkpoint that freezes the portfolio's state, computes P&L attribution, and creates an audit trail. It serves three purposes:

1. **Risk management** — captures the portfolio's MTM, Greeks, and exposure at a known point in time.
2. **P&L attribution** — decomposes today's P&L into components (new trades vs. market moves) for management reporting.
3. **Audit trail** — creates an immutable snapshot that can be reviewed by compliance, model governance, or external auditors.

2 EOD Workflow

2.1 Step 1: Submit EOD

The operator clicks **Submit EOD** on the EOD tab. This triggers a server-side process:

1. Load all open PRS trades from the trade store.
2. Load the current market state (yield curve, hazard term structures, risk-free rate).
3. Revalue every trade using the Delta Engine (fair spread, MTM, FS01, risky annuity).
4. Compute daily P&L by comparing today's MTM against the previous EOD snapshot.
5. Generate the EOD snapshot JSON.
6. Save to `trading/eod/EOD-YYYYMMDD.json`.
7. Optionally generate a PDF report.

2.2 Step 2: Review Snapshot

The EOD tab displays:

- A portfolio summary card (total notional, total MTM, daily P&L, running P&L).
- A position-level table showing each trade's contribution.
- A P&L attribution chart (new trade P&L vs. market P&L).

2.3 Step 3: Download PDF

Each EOD snapshot can be downloaded as a PDF report containing:

- Portfolio summary and key metrics.
- Position detail tables.
- P&L attribution breakdown.
- Market state snapshot (yield curve, hazard rates at close).

3 P&L Attribution

Daily P&L is decomposed into three components:

3.1 New Trade P&L

For trades entered today:

$$\text{P\&L}_{\text{new}} = \frac{s_{\text{fair}} - s_{\text{trade}}}{10,000} \times A \times N \times d$$

This captures the “day 1” P&L from executing at a spread different from fair value. A payer who buys protection at 45 bps when fair is 50 bps has an immediate gain.

3.2 Market P&L

For existing trades (entered before today):

$$\text{P\&L}_{\text{market}} = \text{MTM}_{\text{today}} - \text{MTM}_{\text{yesterday}}$$

This captures the change in value from overnight curve moves. If the hazard rate widened (flood risk increased), payer positions gain and receiver positions lose.

3.3 Running P&L

The cumulative P&L since inception:

$$\text{P\&L}_{\text{running}} = \sum_{\text{all days}} \text{P\&L}_{\text{daily}}$$

3.4 Realized P&L

For closed trades, the final P&L is locked in at close-out:

$$\text{P\&L}_{\text{realized}} = \frac{s_{\text{close}} - s_{\text{trade}}}{10,000} \times A_{\text{close}} \times N \times d$$

4 EOD Snapshot Structure

Each snapshot is a self-contained JSON document:

eod_id Unique identifier (e.g. EOD-20260410).

date ISO date string.

market_state_snapshot Complete yield curve and hazard term structures at market close.

portfolio_summary Aggregate metrics: `num_trades`, `total_notional`, `total_mtm`, `total_daily_pnl`, `total_running_pnl`.

positions Per-trade detail: swap ID, gauge, notional, fair spread, MTM, daily P&L, running P&L.

Snapshots are immutable once written. They are never updated retrospectively. This ensures the audit trail is reliable.

5 P&L Time Series

The EOD tab and Market tab provide three P&L charts:

Daily P&L A line or bar chart showing each day’s total daily P&L. Green bars for positive days, red for negative.

Cumulative Running P&L A line chart tracking the portfolio’s total unrealised P&L over time. The slope indicates whether the book is trending profitably.

Attribution Stacked Bars Each day’s P&L split into `new_trade_pnl` (blue) and `market_pnl` (orange). Shows whether P&L is driven by trading activity or by market moves.

6 Trade Close-Out

6.1 Close-Out Process

A trade is closed by clicking **Close** on the Blotter row and entering a **negotiated close-out spread** in the modal dialog.

The server:

1. Recomputes the risky annuity at the current yield curve and remaining tenor.
2. Calculates the settlement amount:

$$\text{settlement} = \frac{s_{\text{close}} - s_{\text{trade}}}{10,000} \times A \times N \times d$$

3. Determines settlement direction: positive = receivable (the closer receives cash), negative = payable.
4. Sets the settlement date to T+2 (two business days).
5. Updates the trade JSON with close-out metadata.
6. Regenerates the trade PDF with a close-out section.

6.2 Realized P&L

Once closed, the trade’s P&L is “realized” and frozen. It no longer appears in the open blotter but contributes to the portfolio’s total realized P&L in EOD summaries.

7 Parameter Reference

Parameter	Default	Increase	Decrease
DEFAULT_YIELD_CURVE	3.5–4.5%	Changes discount factors at EOD	Changes discount factors

RECOVERY	0.0	Reduces settlement amounts	Full loss (higher settlements)
SPREAD_OFFSET	3–10 bps	Wider bid-ask at inception	Tighter execution
NUM_BUSINESS_DAYS	252	Annualisation denominator	Affects daily rate conversion

Document History

Version	Date	Author	Description
1.0	10-Apr-2026	MKM Research Labs	Initial release